

Andrew R. Noble

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EDUCATION

Carnegie Mellon University

M.S., Robotic Systems Development

Pittsburgh, PA

May 2027

California Polytechnic State University

B.S., Mechanical Engineering (Mechatronics Concentration)

GPA: 3.8/4.0

San Luis Obispo, CA

March 2021

SKILLS

Software: Python, Linux, ROS2, PyTorch, Typescript, HTML, CSS, C, Go, Github CI/CD, PostgreSQL

Hardware: Mechanical CAD, soldering, crimping, electrical schematics

PROFESSIONAL EXPERIENCE

SparkLab

Founder

San Francisco, CA

Jan 2025 - Aug 2025

- Built AI-driven diagram generation tools (Python/TS) for STEM education
- Interviewed 25 teachers, grew to 3000 users, attracted investor interest
- Conducted user research and iterated rapidly through multiple prototypes

Tortuga Agricultural Technologies

Field Robotics Engineer

Denver, CO

Oct 2021 - May 2023

- Wrote ROS2 node in Python that diagnoses end-effector HW, reduces debug time from 30 to 3 mins
- Designed, 3D-printed test stand for ultraviolet light, allowing people to work safely in lab
- Developed ultraviolet fungi treatment and validated it through field trials ([video](#))
- Addressed fleet-wide slip-ring comms failure, enabling two record 1000 kg harvest days
- Performed extensive troubleshooting, bring-up, and repair through entire tech stack

Tortuga Agricultural Technologies

Field Robotics Engineer Intern

Santa Maria, CA

May 2020 - Aug 2020

- Conducted software release and feature tests, communicated results to software team

Coffman Engineers

Mechanical Engineering Intern

Anchorage, AK

May 2019 - Aug 2019

- Developed pipeline loading estimates per environmental constraints on AK's northern oil fields
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PROJECTS & TECHNICAL WORK

State Estimation

- EKF and particle filter implementations for mobile robot localization

Perception & Computer Vision

- Numpy/OpenCV implementations of photometric depth inference, IOU-based tracking, object detection

TinyGPT

- Pretrained a small transformer language model from scratch in raw PyTorch to understand ML fundamentals

Circuit Tutor

- Web application that uses AI to generate electrical circuits for STEM education: circuit-tutor.xyz
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OTHER WORK EXPERIENCE

- REI, Sales
- Alyeska Pipeline Service Company, Drafter

Jan 2024 - July 2024

Aug 2024 - Jan 2025